Team : Troopers

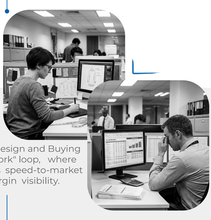
BRIDGING THE GAP

Automated PLM-to-Costing Integration

THE CURRENT LANSCAPE AND THE GAP

While designers successfully upload critical data into the Product Lifecycle Management (PLM) system, a significant bottleneck occurs immediately after, creating a critical gap between design finalization and financial evaluation. Currently, buyers are forced to effectively "start over," manually re-entering that same data to calculate costs, resulting in a labor-intensive and disconnected workflow.

PROBLEM STATEMENT



Manual data silos between Design and Buying create a high-friction "re-work" loop, where redundant data entry delays speed-to-market and obscures real-time margin visibility.

SOLUTION

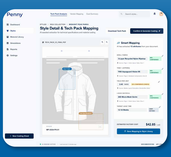
AN AUTOMATED BRIDGE

An automated bridge between the PLM tech pack and a costing engine.
The Concept: An integrated application serving as an automated bridge.
The Connection: Links the PLM Tech Pack directly to a dynamic Costing Engine.
The Mechanism: Eliminates redundant manual tasks while ensuring 100% data integrity.

INTRODUCING

Penny

A professional PLM-to-Costing bridge



CORE FEATURE 1

Automate Data Mapping (AI & OCR)

Direct BOM Extraction:
-Utilizes Natural Language Processing (NLP) and Optical Character Recognition (OCR).
-Pulls fabric types, trim counts, and material weights directly from PLM tech sheets.

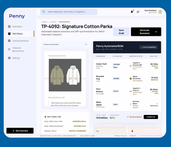
Auto-Populate Unit Costs:
-Linked to centralized Material Library/ERP.
-Instantly applies the most recent "market price" or "contracted price" upon material identification.

CORE FEATURE 2

"Smart" Tentative Costing

Strategic Scenario Generation (Tiered):
Moves beyond random options to generated strategies:
Budget: Lowest cost materials, high-yield layout.
Target: Exactly as specified in the Designer's original Tech Pack.
Premium: Higher-grade trims/finishes, sustainable alternatives.

Historical Benchmarking:
Uses Machine Learning (ML) to analyze past projects.
Suggests baseline costs from similar past garments to ensure "first-time right" accuracy.



CORE FEATURE 3

Real-Time "What-If" Simulations

Dynamic Adjustments:
-Swap a single trim (e.g., metal zipper to nylon).
-See instant impact on Total FOB (Free on Board) price without re-entering data.

Yield Optimization:
-Integrated marker-efficiency predictor.
-Estimates fabric wastage based on garment complexity for accurate "landed cost."

CORE FEATURE 4

System Interoperability & Efficiency Impact

Single Source of Truth:
APIs keep the app synced with PLM. Designer changes trigger notifications or auto-updates to costing.

Supplier Portal Integration:
Secure fields for suppliers to input raw material quotes directly, updating tentative models automatically.

Feature	Manual Process (Current)	Integrated App (Future)
Data Entry	30–60 mins per style	< 5 mins (Automated)
Accuracy	High risk of typos/omissions	High (linked to Library)
Decision Speed	Days (waiting for calcs)	Instant (3 scenarios)
Waste Planning	Estimates/Guesses	Optimized via Yield AI

WORKFLOW

HOW PENNY WORKS

Trigger:
Designer hits "Finalize" on PLM

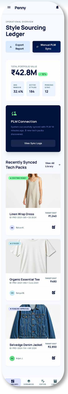
Auto-Sync:
App pulls BOM (Fabric, GSM, Trims, Sketches).

Cross-Reference:
App applies Global Library rates (Market rates, Factory Labor).

Dashboard:
Buyer sees 3 Tentative Cost Cards (Budget, Target, Premium).

Review & Tweak:
Buyer selects option, makes dynamic adjustments (updates FOB instantly).

Export:
One click pushes chosen cost back to PLM for production.



STRATEGIC PRO-TIP

STYLE TEMPLATE

To make this truly seamless, implement "Style Templates." If the Designer tags a sketch as a "Basic Crew Neck Tee," the app should automatically know the standard labor minutes and thread consumption required, leaving the Buyer to only worry about the specific fabric price.

THE ULTIMATE BENEFIT

Faster Speed-to-Market:
Instant costings mean faster acceptance and production starts.

Reduced Administrative Load:
Frees Buyers to focus on strategy and negotiation, not data entry.

Data-Driven Decisions:
Accurate, real-time data ensures profitable margins.

INSTANT | ACCURATE | AUTOMATED

"PENNY FOR YOUR TIME"

Team Members : Sushant | Neena | Harshit